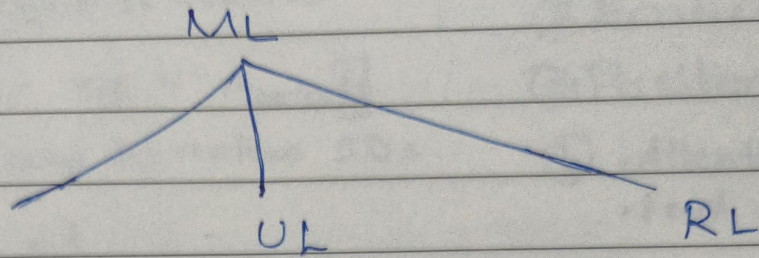


Agentic AI



- XGBoost

SL

UL

RL

- Random forest

- SVM

- KNN

- CAT Boost

- linear regression

- decision trees

$$y = m_1x + m_2x + c$$

x_1	x_2	y
1000	5	500000
2000	10	750000

$$m_1 \times 1000 + m_2 \times 5 = 500000$$

$$m_1 \times 2000 + m_2 \times 10 = 750000$$

Activation functions

- Linear
- Sigmoid
- Tanh
- ReLU
- Leaky ReLU
- Softmax :- $P(y_i) = \frac{e^{x_i}}{\sum_{j=1}^n e^{x_j}}$

Techniques to stop overfitting

- Batch Normalization
- Early Stopping
- Dropout
- Regularization $\leftarrow L_1, L_2$
- Data augmentation
- Cross validation
- Reduce model complexity

Tokens $\Rightarrow$ like a dictionary for NLPs.

LSTM (Long short-term memory)

- Input gate
- Output gate
- Forget gate

$\rightarrow$ attention

I'm going to the river. I am going near the bank.

cat sat on the mat
↓ Text to tokens

["cat" "sat" "on" "the" "mat"]
↓ Tokens to token IDs

[1, 2, 3, 4, 5]
↓ Embedding

[0.25, 0.25, ...] [0.3, ...] [...] [...] [...]

① Tokenization

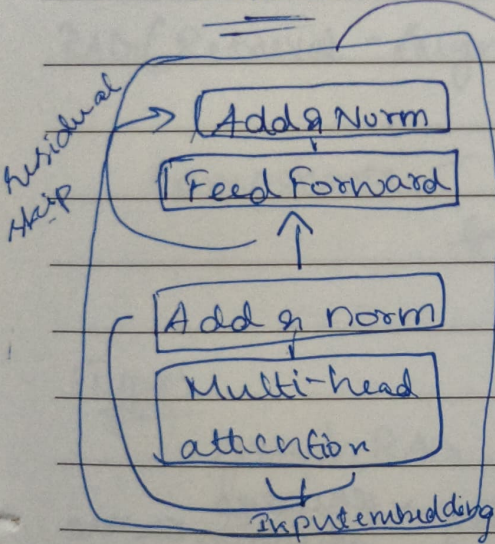
② Embeddings.

③ Positional encoding.

④ • Attention scores
• Feed forward

⑤ Decoder

Encoder



Multihed attention

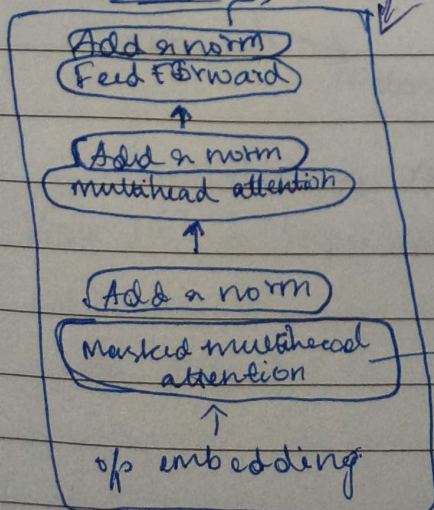
Query(Q) ⇒ What word looking for

Key(K) ⇒ Labels

Value(V) ⇒ actual information

$$\text{Attention} = \text{softmax} \left(\frac{Q K^T}{\sqrt{d_k}} \right) V$$

Decoder → Linear → softmax → o/p probabilities



, masks the future values for training.

$V = [1, 2, 3] \Rightarrow \text{softmax}$

$$\frac{e^{x_1}}{\sum_{j=1}^n e^{x_j}} = \frac{e^{x_1}}{e^{x_1} + e^{x_2} + e^{x_3}} = \frac{2.718}{30.193} = 0.090$$

$$e^{x_1} \Rightarrow 2.718$$

$$e^2 \Rightarrow 7.389$$

$$e^3 \Rightarrow 20.086$$

$$\underline{\underline{30.193}}$$

$$\frac{7.389}{30.193} = 0.245$$

$$\frac{20.086}{30.193} = 0.665$$

$$0.090 + 0.245 + 0.665 = 1 \quad (\text{it should be 1})$$

Training an LLM

① Data

② Compute/Infer

③ Architecture

④ Evaluation

⑤ Objective

Text $\rightarrow$ Tokens $\rightarrow$ Embeddings $\rightarrow$ Positional encoding $\rightarrow$ encodes attention scores

Softmax $\leftarrow$ linear transformation $\leftarrow$ decoder

ChatGPT, Claude $\Rightarrow$ Decoder only transformer architecture
GPT $\Rightarrow$ BERT (Bidirectional Encoder Transformer).

Tuning - Knowledge cutoff

- one shot prompt
- few shot prompt

Limitations

- Overfitting
- Underfitting
- Forgets prev response
- Hallucination

weights of low rank matrix are only changed

Finetuning - Information extraction

- Supervised FT
 - Parameter efficient FT
 - LoRA (Low-Rank Adaption)
 - QLoRA (Quantized LoRA)
- eg: llama - 34 billion parameters → 4 bit/shits

RAG (Retrieval - Augmented Generation)

Components

- Retriever - retrieves information
- Generator - a decoder/generate response based on the query.

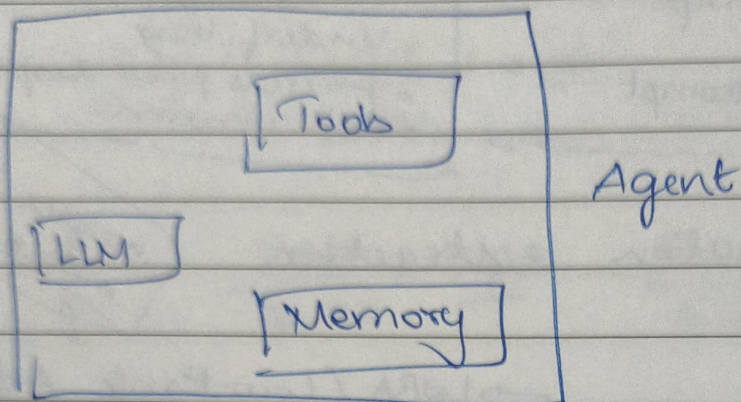
Types.

- Graph RAG
- Hybrid RAG
- Naive RAG

Human-in-the-loop

An agent is a LLM with the tools (functions).

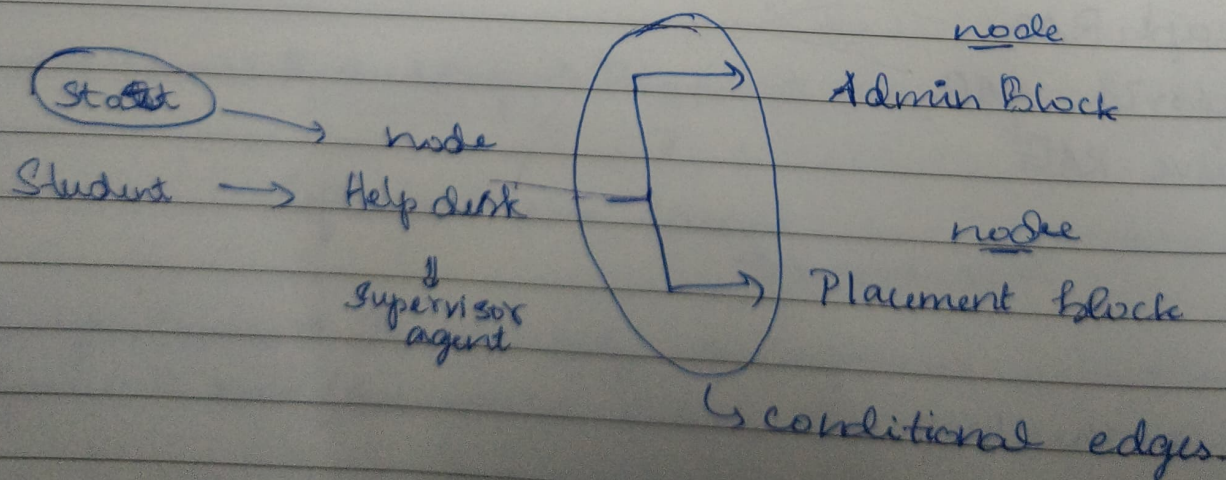
Every agent should have a memory (what you learn from what you did)



Tools

- Extract student info
- Analyse reports
- Draft mails
- Verify errors.

→ You should not get an "argument error".



State → memory that a supervisor agent can use to know what has done till now.

